ABM Assignment 1 - Smart Food Delivery Agents: deliverables

Status: complete. Open items: Section, Instructor's full name and Group members are blank on the title page.

Files

- ABM_Assignment1_SherAliSaleem.zip: everything in one folder tree (use this one)
  - report/ (PDF and DOCX)
  - food_delivery_agents/ (code, diagrams/, results/)
  - dashboard/ (one HTML file)
  - README.md
- Assignment1_Agent_Based_Modeling_SherAliSaleem.pdf (30 pages, A4)
- Assignment1_Agent_Based_Modeling_SherAliSaleem.docx (same content, editable)
- food_delivery_agents.zip (code, 7 diagrams, results only)
- Same report as an online Docs page (Claude Docs)

Dashboard (extra, not required by the brief)

- Interactive page "Smart Food Delivery Agents", published as a private Claude artifact: https://claude.ai/artifact/DBvxmyX8PxPyh4Xkf2nbRL
- Also saved as smart_food_delivery_dashboard.html (works offline in any browser)
- Lab tab: load Test 1-3 or a random situation, or set weather, time, customer, food, deadline, traffic, feed and alerts; all 7 agents decide side by side, each with its own working and its architecture diagram
- Scoreboard tab: the 1,000-delivery benchmark by kind of delivery (on time, average time, average cost)
- Learning tab: replay the learning agent's 600 training deliveries
- The page runs a JavaScript copy of the Python agents; it matched the Python output on 67,803 situations (290,047 checks, 0 differences). Benchmark and learning figures come from the Python code.
- Desktop-first: the user asked for it to look good on the web; phone width works but was not polished

Contents of the report

- Title page, contents, overview, simulated world
- Per agent (1-7): behaviour, architecture diagram, full code
- 3 test cases with tables, comparison, conclusion
- Shared code, appendices (raw output, decision log, benchmark)

Run

cd food_delivery_agents python run_tests.py python benchmark.py

Test results

- Test 1 (normal): all 7 chose A, 14.0 min

- Test 2 (heavy, several routes): B best, 15.2 min
- Test 2: Simple Reflex took C (27.5 min); Goal-Based took A (29.8 min, 0.2 min spare)
- Test 3 (heavy + rain + urgent, 20 min deadline): only C is on time (18.8 min)
- Test 3 on time: Goal-Based, Learning, Contextual
- Test 3 late (B, 21.4 min): Simple Reflex, Model-Based, Utility, Hierarchical

Benchmark (1,000 random deliveries, 307 urgent)

- Contextual: 96.5% on time, 92.2% urgent
- Hindsight oracle: 96.8% on time, 92.2% urgent
- Learning (trained): 95.0% / 87.6%
- Hierarchical 94.5, Model-Based 94.4, Utility 94.0, Simple Reflex 93.2, Goal-Based 88.0

Conclusion

The Contextual Agent is the most suitable: it uses rain, alerts, rush hour, VIP and hot-food context that the traffic feed does not show.

Viva notes

- The brief says "six" agents but lists seven; all seven were built.
- The simulator is self-designed, so results show how each architecture uses information, not real-world performance.
- Hidden effects: route B in rain x1.40 (flooded underpass), route A in heavy traffic x1.30.
- Learning agent learned B-in-rain from 600 simulated past deliveries.
- Hierarchical agent's 10% safety buffer ruled out route C in Test 3.
- Goal-Based is a satisficer: first route that passes the deadline test.